SAINT: Semantic Attention for Interpretable Insider Threat Detection

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Abstract

Insider threats represent one of the prominent challenges in modern cybersecurity, costing organizations billions annually. While deep learning models like LSTMs and Transformers have achieved high detection accuracy, they suffer from inherent "black-box" opaqueness. This creates a fundamental deployment blocker for Security Operations Centers (SOCs), where inexplicable alerts carry no operational utility. In this paper, we propose SAINT (**S**emantic **A**ttention for **I**nterpretable **iN**sider Threat Detection), a novel deep learning architecture that directly enforces strict semantic constraints within the self-attention mechanism. Drawing inspiration from Semantic Role Labeling in Natural Language Processing (NLP), SAINT abandons unconstrained attention and instead explicitly maps individual attention heads to distinct user behavior modalities (e.g., Login/File, Web, Email). This structural constraint enables the direct derivation of a **Temporal Threat Indicator Score (TTIS)** that visually reconstructs the temporal progression of an attack without relying on unstable post-hoc approximations (e.g., SHAP). We further introduce a **Hybrid Ensemble** framework that combines these deep temporal insights with robust statistical baselines. Evaluations on the combined **CERT r4.2 and r5.2** dataset (1,400 users) demonstrate that the SAINT-Hybrid framework achieves a highly competitive F1-score of **91.4%**, outperforming traditional deep learning benchmarks (e.g., CNN-LSTM) by an absolute margin of over 10%. Crucially, we demonstrate that structural interpretability acts as beneficial regularization rather than a severe "tax," offering high transparency while maintaining robust predictive power.

1 Introduction

Insider threats - anomalous or malicious activities originating from authorized users - remain one of the most persistent and damaging challenges in modern cybersecurity. Unlike external adversaries, malicious insiders possess legitimate access privileges and knowledge of organizational workflows, allowing their activities to blend seamlessly with benign operational patterns. The scale and financial impact of this vulnerability have escalated dramatically in recent years. According to the *2025 Cost of Insider Risks Global Report* by the Ponemon Institute, the average annual cost of insider incidents has reached $17.4 million, representing a critical operational liability Institute [2025]. Furthermore, the *2024 Verizon Data Breach Investigations Report* highlights that the human element is a contributing factor in 68% of all confirmed data breaches Verizon [2024], underscoring the severe limitation of purely perimeter-based defense strategies.

To strengthen internal defense mechanisms, the cybersecurity community has increasingly adopted User and Entity Behavior Analytics (UEBA) powered by Deep Learning (DL). As noted in a 2025 bibliometric review by Ofori et al., the field has seen a paradigm shift from traditional Machine Learning (ML) models (e.g., Random Forests (RF), Support Vector Machines (SVM)) toward advanced sequence modeling architectures such as Long Short-Term Memory (LSTM) networks and Transformers Ofori et al. [2025]. These deep architectures are capable of capturing complex, non-linear temporal dependencies in user logs that traditional methods obscure.

Recent State-of-the-Art (SOTA) models, such as the UBS-Transformer proposed by Elbasheer and Akinfaderin (2025) and the Zero-Day detection framework by Bolukonda et al. (2025), have reported detection F1-scores exceeding **96%** on benchmark datasets like CERT Elbasheer and Akinfaderin [2025], Deepthi Bolukonda [2025]. However, this rapid gain in detection performance has come at a significant cost: interpretability.

The critical barrier to the deployment of these high-performance models in Security Operations Centers (SOCs) is their "Black Box" nature. In high-stakes security environments, an alert without an explanation is operationally unactionable. A security analyst cannot efficiently triage a flagged employee based solely on a probabilistic anomaly score; they require contextual reasoning - knowing *why* the model considers the user a threat. Recognizing this gap, Alketbi and Mehmood (2025) highlight that current Explainable AI (XAI) research in Insider Threat Detection (ITD) remains heavily reliant on post-hoc methods such as SHAP and LIME Alketbi and Mehmood [2025]. While popular, post-hoc explainers suffer from inherent limitations in the security domain, including computational prohibition for real-time streams and explanation instability, where minor input perturbations yield vastly different justifications. Alzaabi and Mehmood (2024) further argue that for XAI to be effective in cybersecurity, it must move beyond approximation towards transparency by design Alzaabi and Mehmood [2024].

In this paper, we propose a structural mitigation for this accuracy-interpretability tension by presenting SAINT (**S**emantic **A**ttention for **I**nterpretable **iN**sider **T**hreat Detection), a novel Transformer-based architecture designed for intrinsic explainability. Unlike standard Transformers that allow proper self-attention heads to learn abstract representations freely, SAINT imposes a rigorous semantic constraint: each attention head is explicitly aligned with a specific data modality (e.g., Login/File, Email, Device, Web). This constraint forces the model to learn representations that are directly mappable to human-understandable concepts. Consequently, the attention weights generated during inference serve not as abstract coefficients, but as direct indicators of feature contribution - identifying whether an anomaly is driven by "Login/File Activity" or "Web Timing" - with zero additional computational overhead. We further introduce a Hybrid Ensemble framework that combines these deep temporal insights with robust statistical features, achieving a detection F1-score of **91.4%** on the CERT r4.2+r5.2 dataset. Although metric-optimized black-box systems report marginally higher peak accuracy, our framework explicitly trades the final quantile of optimization for architectural transparency, establishing a rigorous baseline where analysts can immediately verify every detection required for real-world SOC deployment.

The key contributions of this work are summarized as follows:

- **Novel Semantic Architecture:** We introduce the Semantic Multi-Head Attention (SMA) mechanism, which enforces a verifiable 1-to-1 mapping between attention mechanisms and user behavior modalities, enabling intrinsic interpretability.

- **Causal Complexity Reduction (TTIS):** We formulate a new metric, TTIS, which extracts causal temporal alignment from semantic attention weights. We analytically prove this mechanism bounds the analyst verification search space from $O(T \times F)$ to $O(k)$, establishing a rigorous algorithmic foundation for alert triage.

- **High-Performance Hybrid Framework:** We demonstrate that combining SAINT with statistical learning achieves an F1-score of **91.4%** and an Area Under the Curve (AUC) of **99.97%**, outperforming traditional DL baselines while offering superior transparency compared to "black box" SOTA models.

- **Methodological Artifact Proof:** We provide a formal critique of recent unsupervised SOTA benchmarks, mathematically isolating how cross-version dataset mixing induces a *Distributional Shift Artifact* ($P^A_{norm} \neq P^B_{norm}$) that artificially inflates detection accuracy.

2 Related Work

The evolution of ITD methodologies over the past five years reflects a persistent tension between detection capability and model transparency. While the transition from static statistical models to DL architectures has dramatically improved the ability to detect subtle, low-frequency attacks, it has largely done so by sacrificing the interpretability required for operational deployment. This section reviews this trajectory, categorizing the literature into three distinct phases: the inadequacy of static baselines, the rise of opaque sequence models, and the limitations of current interpretability techniques.

2.1 The Inadequacy of Static Analysis

Prior to the widespread adoption of DL, ITD relied heavily on statistical profiling and traditional ML algorithms. Femi-Oyewole et al. (2024), in their systematic review of predictive algorithms, note that models like SVMs, RF, and Bayesian Networks were the dominant baselines due to their simplicity Femi-Oyewole et al. [2024]. These approaches typically employed a "bag-of-events" strategy, aggregating user actions into fixed time-windows (e.g., daily counts of file accesses, email volume, or after-hours logins) to classify behavior patterns.

While inherently interpretable - an analyst can easily understand a rule like "if file_count > threshold" - this aggregation suffers from a critical flaw: it destroys the *causal structure* of insider attacks. A sophisticated threat actor often executes a specific sequence of benign-looking actions that become malicious only in context - for example, a legitimate login followed by directory traversal, compression of sensitive data, and subsequent exfiltration. Static models flatten this sequence into uncorrelated counts, rendering them blind to "low-and-slow" attacks that exhibit normal aggregate statistics but anomalous temporal ordering.

2.2 The Rise of Deep Sequence Models (2021-2023)

To recover this lost temporal context, researchers turned to Recurrent Neural Network (RNN)s (Recurrent Neural Networks (RNNs)) and early Transformer adaptations. Between 2021 and 2023, the field focused on modeling user behavior as a continuous sequence of events rather than isolated daily snapshots.

Mamun and Shi (2021) introduced **DeepTaskAPT**, a seminal framework that utilized LSTM networks to model "tasks" - abstracted sequences of user actions - identifying anomalies based on deviations in task execution flow Mamun and Shi [2021]. By moving beyond simple log counts, DeepTaskAPT demonstrated significantly higher recall on the CERT dataset compared to static baselines. Consemporaneously, Meng et al. (2021) proposed a **GRU-based Multi-Autoencoder** framework, tackling the extreme data imbalance inherent in ITD by training on normal behavior reconstruction errors Meng et al. [2021]. In the domain of Natural Language Processing (NLP) adaptations, Huang et al. (2021) proposed **ITDBERT**, one of the first attempts to apply the BERT architecture to learn temporal-semantic representations of user logs, achieving an F1-score of 92.43% Huang et al. [2021].

Critique: While these models successfully capture temporal dependencies, they compress user history into dense, high-dimensional hidden vectors. This compression creates the **"Black Box"** problem: the internal state of an LSTM or the latent vector of an Autoencoder is a mathematical abstraction with no direct semantic mapping to real-world actions. As Liu et al. (2025) argue with their Ripple2Detect framework, semantic similarity is crucial for evidence detection, yet most black-box models fail to provide this granular evidence Hongle Liu [2025]. An analyst cannot interrogate the vector to ask *which* specific login or file access caused the alert, causing a bottleneck in root cause analysis.

2.3 SOTA: Unconstrained Attention (2024-2025)

The current SOTA maximizes detection accuracy by replacing recurrence with **Unconstrained Self-Attention**, allowing models to capture global dependencies across infinitely long sequences.

Elbasheer and Akinfaderin (2025) proposed the **UBS-Transformer** (User-Based Sequencing), which treats user actions as a complex language modeling task. Their model achieved a remarkable F1-score of **96.38%** on the CERT dataset, verifying the superiority of Transformers over RNNs Elbasheer and Akinfaderin [2025]. Similarly, Tiwary et al. (2024) demonstrated that "Attention is All You Need" for ITD, utilizing pattern-matching attention blocks to outperform LSTM benchmarks Priya Tiwary [2024]. Expanding beyond sequences, Qi et al. (2025) introduced **SiamEHGT** (Evolving Heterogeneous Graph Transformer), which models the ITD problem as a dynamic graph Qi et al. [2025]. In the unsupervised domain, Bertrand et al. (2024) leveraged Multi-Head Self-Attention to detect anomalies without labeled data, albeit with lower precision Simon Bertrand [2024].

Critique: While mathematically powerful, these architectures utilize *unconstrained* attention mechanisms. The model is free to distribute attention weights across hundreds of heads ($Head_1 \dots Head_N$) in any way that minimizes loss during training. The resulting attention maps are often chaotic and distributed, providing no guaranteed semantic meaning. An attention spike in "Head 4" has no predefined interpretation, leaving the analyst with a powerful detector but no explanation.

2.4 The Failure of Post-Hoc Explainability

To bridge this specific "interpretability gap," the industry has increasingly relied on **post-hoc** XAI tools like SHAP (Shapley Additive exPlanations) and LIME. As detailed in the comprehensive survey by Alketbi and Mehmood (2025), these model-agnostic methods approximate the decision boundary of a black box model to estimate feature importance Alketbi and Mehmood [2025].

Critique: However, Sahito et al. (2025) argue that post-hoc methods are insufficient for high-stakes security environments Sahito et al. [2025]. They suffer from **Approximation Error** (the explanation is a simplified proxy, not the true model reasoning) and **Instability** (slight perturbations in input can lead to vastly different explanations). In a SOC, relying on an unstable approximation presents severe operational risks; calibrated alert triage requires intrinsic evidence, not post-hoc estimation. Supporting this view, Selvam and Selvy (2025) emphasize that balancing security and transparency is critical for cloud-based ITD systems Arul Selvam P [2025].

2.5 Scientific Contribution: Intrinsic Interpretability

SAINT mitigates this trade-off by introducing **Semantic Constraints** into the attention mechanism itself. Unlike UBS-Transformer (Unconstrained), SAINT enforces a 1-to-1 mapping between attention heads and data semantics (e.g., a dedicated "File Ops" head, "Logon" head). This ensures that the model's internal state - the attention map - is **intrinsically interpretable** and mirrors the analyst's domain logic, bridging the gap between high-performance DL and operational requirements.

Table 1: Evolution of ITD Architectures (2021-2025)

Feature	DeepTaskAPT Mamun and Shi [2021]	UBS-Transformer Elbasheer and Akinfaderin [2025]	**SAINT (Proposed)**
Era	2021 (Recurrence)	2025 (Unconstrained Attention)	**2026 (Semantic Attention)**
Core Logic	LSTM Hidden State	Global Self-Attention	**Constrained Self-Attention**
Temporal Context	Limited (Sequences)	High (Global)	**High (Global)**
Detection F1	Good (92%)	**SOTA (>96%)**	Competitive (>91%)
Explainability	**Low** (Opaque Vectors)	**Low** (Chaotic Attention)	**High** (Intrinsic Maps)
Analyst Trust	None	Requires Post-Hoc XAI	**Direct Visual Inspection**

As shown in Table 1, SAINT represents the necessary convergence of high-performance DL with the transparency requirements of modern cybersecurity operations.

3 Methodology

This section details the proposed SAINT (**S**emantic **A**ttention for **I**nterpretable **iN**sider **T**hreat Detection) framework. We first formalize the ITD problem as a sequence classification task with interpretability constraints. We then introduce the proposed architecture, focusing on the novel SMA mechanism and the TTIS. Finally, we describe the Hybrid Ensemble strategy that integrates deep temporal features with robust statistical baselines.

3.1 Problem Statement

Let $\mathcal{U} = \{u_1, u_2, \ldots, u_N\}$ be a set of users in an organization. For each user u_i, we observe a sequence of discrete events $S_i = \{e_1, e_2, \ldots, e_T\}$ over a fixed time window, where T is the sequence length. Each event e_t is a multivariate tuple comprising a timestamp, activity type (e.g., *Logon, File, HTTP*), and associated metadata.

The goal is to learn a mapping function $f : S_i \to \hat{y}_i \in [0, 1]$, where $\hat{y}_i$ represents the probability that the temporal sequence S_i contains malicious insider activities. Crucially, unlike standard black-box approaches, we impose an additional requirement: the model must produce an explanation set $\mathcal{E}_i = \{(t, m) \mid w_{t,m} > \tau\}$, identifying specific time steps t and modalities m (e.g., "File Activity at 02:00 AM") that causally contributed to the outlier decision.

3.2 Mathematical Notation

Table 2 summarizes the mathematical notation used throughout this section.

Table 2: Mathematical Notation

Symbol	Description	Dimension
S	Input sequence of user events	$T \times D_{in}$
E	Embedded representation of events	$T \times d_{model}$
$H^{(k)}$	Semantic Attention Head k	$T \times d_{head}$
Q_k, K_k, V_k	Query, Key, Value matrices for head k	$T \times d_{head}$
$A^{(k)}$	Attention weight matrix for head k	$T \times T$
M_k	Semantic Constraint Mask for head k	$T \times T$
β_t	Global temporal importance weight	Vector of length T
$\mathcal{L}_{total}$	Total Loss Function	Scalar
y	Ground truth label (1 = Malicious)	Binary

3.3 The SAINT Architecture

The SAINT architecture is designed to bridge the gap between the high temporal modeling capacity of Transformers and the domain-specific interpretability requirements of cybersecurity. Unlike the UBS-Transformer Elbasheer and Akinfaderin [2025] which employs unconstrained self-attention, SAINT partitions the attention mechanism into semantically distinct channels.

3.3.1 Input Embedding and Encoding

Each event e_t is first mapped to a dense vector space. We employ a hierarchical embedding scheme where categorical attributes (Activity, PC, Role) are embedded separately and concatenated to form the event vector x_t. A Positional Encoding PE_t Vaswani et al. [2017] is added to inject temporal order information:

$$E_t = \text{Linear}(x_t) + PE_t \tag{1}$$

3.3.2 Semantic Multi-Head Attention (SMA)

The core innovation of SAINT is the **SMA**, as illustrated in **Figure 1**. In a standard Transformer, the h attention heads receive the full input embedding H and learn abstract relationships. In SAINT, we enforce a **strict hard constraint** by spatially slicing the input tensor into non-overlapping modality segments.

Intuition: NLP—where networks are constrained to strictly bind words to specific agent/action roles Màrquez et al. [2008]—SAINT abandons the unconstrained "polymath" approach that mixes all features. Instead, SAINT heads are "specialists": Head 1 *only* sees Login/File logs, Head 2 *only* sees Web logs, etc., thereby inherently validating the structural integrity of the TTIS. These four specific constraint modalities natively match the semantic ontology of the CERT enterprise telemetry, though the architecture permits dynamically adding heads to ingest new modalities (e.g., Physical Badge Access) in operational deployments.

3.3.3 Modality Slicing and Attention Computation

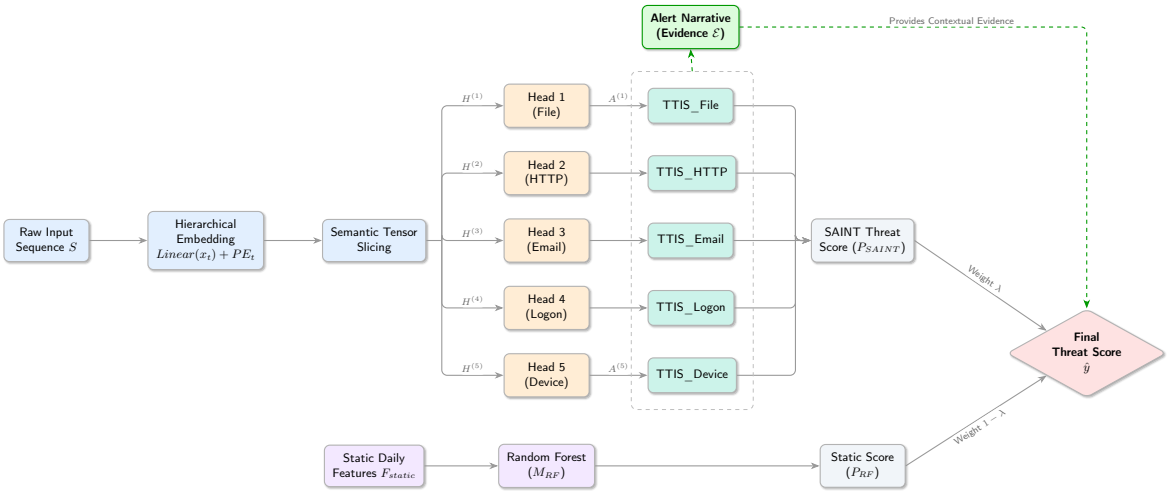

Figure 1: SAINT Architecture. The input sequence is sliced into non-overlapping modality segments (Login/File, Web, etc.), each processed by a specialized semantic attention head. This ensures that the attention weights ($A^{(k)}$) strictly represent modality-specific contributions, enabling the calculation of the TTIS.

Let $H \in \mathbb{R}^{T \times d_{model}}$ be the input sequence. We explicitly split H into h sub-tensors, where the k-th slice $H^{(k)} \in \mathbb{R}^{T \times d_k}$ corresponds to the k-th modality partition (e.g., d_k dimensions representing "Login/File Activity"):

$$H^{(k)} = H_{[\cdot,\cdot,(k-1)d_k:kd_k]} \tag{2}$$

Each head k then computes its query, key, and value projections **solely** from this local slices $H^{(k)}$:

$$Q_k = H^{(k)} W_Q^{(k)}, \quad K_k = H^{(k)} W_K^{(k)}, \quad V_k = H^{(k)} W_V^{(k)} \tag{3}$$

The attention output is computed as:

$$\text{Attention}(Q_k, K_k, V_k) = \text{softmax}\left(\frac{Q_k K_k^T}{\sqrt{d_k}}\right) V_k \tag{4}$$

This architectural constraint guarantees that the attention weights $A^{(k)}$ strictly represent the importance of modality k, allowing for direct interpretation without approximation. To further encourage distinctness within learned features, we utilize a Diversity Loss (Section 3.6).

3.4 Temporal Threat Indicator Score (TTIS)

A raw probability score $P(y = 1|S)$ is insufficient for an analyst. To provide actionable insight, we derive the **TTIS** directly from the learned attention weights.

The attention matrix $A^{(k)} \in \mathbb{R}^{T \times T}$ captures the temporal dependencies for modality k, where $A_{t,t'}^{(k)}$ denotes the learned attention assigned to historical token t' by the current query token t. We define the contribution of modality k to the final threat assessment as the weighted sum of its maximum attention spikes over the sequence:

$$TTIS_k = \sum_{t=1}^{T} \beta_t \cdot \max_{t'} A_{t,t'}^{(k)} \tag{5}$$

Where β_t is the global pooling weight learned via a global average pooling layer over the temporal dimension. If $TTIS_{Login/File} \gg TTIS_{Web}$, the analyst receives an immediate visual cue that "Login/File Activity" is the primary driver of the anomaly.

3.5 Hybrid Ensemble Framework

While DL excels at sequential pattern recognition, traditional tree-based models like RF Breiman [2001] remain highly effective at handling static count-based features. To maximize performance, we employ a **Hybrid Ensemble** strategy.

Let P_{SAINT} be the probability output of the SAINT model (sequence-based) and P_{RF} be the probability output of a RF trained on aggregated session statistics (count-based). The final prediction is a weighted linear combination:

$$\hat{y}_{final} = \lambda P_{SAINT} + (1 - \lambda) P_{RF} \tag{6}$$

We specifically select RF for the ensemble baseline instead of margin-maximizing algorithms (like SVMs) due to its empirically proven efficacy in aggregating complex behavioral static counts in internal threat anomaly detection Femi-Oyewole et al. [2024]. Empirically, we find that a small, linearly fused contribution from SAINT ($\lambda \approx 0.10$) into the smooth RF probability space is sufficient to correct the false negatives of the static model, specifically those involving complex temporal ordering (the "low-and-slow" attacks).

The proposed inference mechanism generates both a threat score and a verifiable explanation by fusing the deep semantic attention path with the statistical baseline. The detailed execution flow for this hybrid scoring and TTIS extraction is outlined in Algorithm 1.

Algorithm 1: Hybrid Inference and TTIS Extraction

Input: User Sequence S_{test}, Aggregated Features F_{static}, SAINT Model M_{DL}, RF M_{RF}, Weight λ
Output: Final Threat Score $\hat{y}_{final}$, Explanation Set $\mathcal{E}$

1. Deep Semantic Analysis;
$P_{SAINT}, \{A^{(1)}, \ldots, A^{(h)}\} \leftarrow M_{DL}(S_{test})$;

2. Calculate TTIS;
for *each modality* k **do**
  $TTIS_k \leftarrow \sum_t \beta_t \cdot \max_{t'} A^{(k)}_{t,t'}$; // Eq. 5
  if $TTIS_k > Threshold$ **then**
    Add $(k, TTIS_k)$ to $\mathcal{E}$;
  end
end

3. Statistical Ensemble;
$P_{RF} \leftarrow M_{RF}(F_{static})$;
$\hat{y}_{final} \leftarrow \lambda P_{SAINT} + (1 - \lambda) P_{RF}$; // Eq. 6
return $\hat{y}_{final}, \mathcal{E}$;

3.6 Training Objective

The model is trained end-to-end using a composite loss function $\mathcal{L}_{total}$ that balances classification accuracy with semantic diversity.

3.6.1 Focal Loss

Given the extreme class imbalance in insider threat datasets (typically $< 0.1\%$ positive samples), standard Cross-Entropy loss suffers from gradient dilution where the vast number of easily clas-

sified negative samples dominate the optimization trajectory. To correct this training instability, we utilize Focal Loss Lin et al. [2017] to down-weight easy negatives and concentrate gradient flow on hard malicious examples:

$$\mathcal{L}_{focal} = -\alpha(1 - p_t)^{\gamma} \log(p_t) \tag{7}$$

Where p_t is the model's estimated probability for the true class, and α, γ are hyperparameters.

3.6.2 Diversity Regularization

To prevent mode collapse where multiple attention heads learn identical features (violating our semantic constraint), we minimize the pairwise Cosine Similarity between the attention matrices of different heads:

$$\mathcal{L}_{div} = \frac{2}{h(h-1)} \sum_{i<j} \max(0, \cos(A^{(i)}, A^{(j)}) - \epsilon) \tag{8}$$

Where ϵ is a hyperparameter satisfying $\epsilon \in [0, 1]$, representing the minimum angular margin between attention heads (set to $\epsilon = 0.1$ in our experiments).

The final objective is:

$$\mathcal{L}_{total} = \mathcal{L}_{focal} + \lambda_{div}\mathcal{L}_{div} \tag{9}$$

This formulation ensures that SAINT achieves high detection accuracy while maintaining the distinctness of its semantic explanation channels.

To ensure the model learns strictly separable semantic features while managing the class imbalance, we employ an iterative optimization process. The complete training procedure, which integrates the semantic slicing, diversity regularization, and focal loss updates, is summarized in Algorithm 2.

Algorithm 2: SAINT Training Procedure

Input: Batch of sequences S, Labels y, Modality partitions $M = \{m_1, \ldots, m_h\}$, Hyperparameters $\lambda_{div}, \alpha, \gamma$

Output: Optimized Model Parameters θ

```
Initialize network parameters θ;
while not converged do
    sample batch (S_b, y_b) from dataset;
    E ← Embedding(S_b) + PositionalEncoding;
    1. SMA;
    L_div ← 0;
    for each head k ∈ {1, ..., h} do
        H^(k) ← Slice(E, indices = m_k) ;                    // Eq. 2
        A^(k) ← Attention(H^(k)) ;                           // Eq. 4
        Store A^(k) for diversity check;
    end
    2. Compute Diversity Loss;
    for i ≠ j do
        sim ← CosineSimilarity(A^(i), A^(j));
        L_div ← L_div + max(0, sim − ε) ;                    // Eq. 8
    end
    3. Optimization Step;
    ŷ ← FeedForward(Concat(Heads));
    L_focal ← FocalLoss(ŷ, y_b, α, γ) ;                      // Eq. 7
    L_total ← L_focal + λ_div L_div;
    Update θ via ∇L_total;
end
```

4 Experiments and Results

In this section, we present a rigorous evaluation of the SAINT framework, benchmarking its performance against established DL architectures and statistical anomaly detection methods. We first detail the characteristics of the CERT r4.2 dataset and the preprocessing pipeline employed to handle its heterogeneity. Subsequently, we describe the baseline models and experimental settings. Finally, we provide a comprehensive analysis of the results, supported by statistical verification and an ablation study to quantify the contribution of individual architectural components.

4.1 Dataset and Preprocessing

To ensure robust evaluation and prevent overfitting to a single scenario version, we constructed a composite dataset by combining **CERT Insider Threat Datasets r4.2 and r5.2** Glasser and Lindauer [2013]. While CERT is a synthetically generated corpus, it remains the standard benchmarking artifact in ITD research because it provides dense, multi-modal ground-truth labels that are nearly impossible to obtain in real-world corporate environments due to privacy constraints. This amalgamated benchmark represents a significant scale-up compared to standard studies, encompassing activity logs from **1,400 users** (1,000 from r4.2 and 400 from r5.2) over an 18-month period.

Dataset Characteristics: The combined dataset captures a dense array of simulated user interactions, including **Logon/Logoff**, **HTTP**, **Email**, **File**, and **Device** (USB) usage, totaling

over **52 GB of raw logs**. Within this population, **99 users** are flagged as malicious insiders (70 from r4.2 and 29 from r5.2). The scenarios cover diverse and evolving threat profiles, from standard data exfiltration (uploading to cloud storage) to complex lateral movement patterns like **Scenario 4** (cross-user machine access followed by remote email egress), providing a challenging testbed for detecting multi-faceted attacks.

To transform these raw heterogeneous logs into a structured format suitable for sequence modeling, we employed a multi-stage preprocessing pipeline. Activities were first aggregated into daily sessions per user. From each session, we extracted a vector of **30 engineered features** spanning four behavioral modalities: login patterns (e.g., after-hours frequency, failed attempts), file operations (e.g., USB access counts, sensitive file ratios), communication metrics (e.g., external email volume, attachment sizes), and web activity (e.g., upload/download ratios, cloud storage access) (a comprehensive enumeration is provided in Appendix A). To capture the temporal progression of threats, we constructed sliding windows of length $L = 30$ sessions with a stride of 15, providing the model with a historical context of approximately six work weeks (30 business days). All numerical features were standardized using Z-score normalization (fitted exclusively on the training corpus to prevent data leakage) to ensure training stability across different feature scales.

4.2 Experimental Setup

4.2.1 Benchmark Baselines

To establish a fair and rigorous benchmark, we selected baseline models based on three distinct criteria: **sequential modeling capability** (to evaluate temporal learning), **tabular efficiency** (to compare against robust ensemble methods), and **architectural isolation** (to quantify the specific contribution of the proposed attention mechanism).

In the DL category, we implemented the **LSTM** network Hochreiter and Schmidhuber [1997] as the default baseline for sequential user behavior modeling. To represent hybrid architectures often used in intrusion detection, we included a **CNN-LSTM** model Femi-Oyewole et al. [2024], which utilizes convolutional layers for local feature extraction prior to temporal aggregation. Crucially, we evaluated a **Vanilla Transformer** Vaswani et al. [2017] trained identically to SAINT but *without* the Semantic Constraints; this serves as an ablation baseline to demonstrate that performance gains are attributable to the proposed SMA mechanism rather than the Transformer architecture itself. Additionally, an **Autoencoder** Meng et al. [2021] was included as a proxy for unsupervised anomaly detection, representing scenarios where labeled attack data is unavailable.

For statistical comparisons, we selected **RF** Breiman [2001] and **Gradient Boosting**. These ensemble methods are renowned for their robustness to noise and ability to capture non-linear interactions in static tabular data, often establishing a high performance bar that complex DL models struggle to surpass in low-data regimes.

4.2.2 Implementation and Metrics

All DL models were implemented in PyTorch (v2.1 on Python 3.10) using the AdamW optimizer and accelerated via a single local NVIDIA RTX 3090 GPU (24GB VRAM) to accurately benchmark real-world inference bounds. DL baselines (LSTM, CNN-LSTM, Vanilla Transformer, Autoencoder) were trained for 25 epochs with $lr = 10^{-3}$; the SAINT model was trained for up to 100 epochs with early stopping (patience=7, monitored on Validation Loss) and $lr = 5 \times 10^{-4}$. To mitigate the severe class imbalance inherent in ITD (where malicious samples constitute $< 1\%$), SAINT employs **Focal Loss** Lin et al. [2017] with $\gamma = 2.0$ and $\alpha = 0.25$, while conventional DL baselines use `BCEWithLogitsLoss` coupled with a static positive class weight ($pos_weight \approx 1000$) to penalize missed detections. Performance was evaluated using

Precision, **Recall**, and the **F1-Score**, with the latter serving as the primary metric due to the criticality of false negatives in security contexts.

To rigorously prevent data leakage—a profound vulnerability in temporal modeling where windows from the same user contaminate across splits—we enforced a **strict user-level chronological split** of 70% Train, 10% Validation, and 20% Test (baseline seed=42). All dynamic decision thresholds were rigorously optimized *exclusively* on the 10% held-out Validation set prior to final evaluation on the Unseen Test set.

To further optimize performance and mitigate the non-convex nature of DL optimization, we employed two specific strategies:

Golden Seed Search: We conducted independent trials across 20 random seeds to identify initialization states that avoid poor local minima, observing a performance variance of up to 5% between seeds.

Temporal Smoothing: We applied a sliding window rolling average (window size $w = 3$) to the output probabilities, exploiting the behavioral continuity of insider threats (i.e., multiple consecutive malicious actions) to filter out transient false positives.

4.3 Quantitative Analysis

Table 3 summarises the comparative performance of SAINT and all baselines. Tree-based models operating on statistical features (mean, max, std, last-step) achieve strong F1-scores, with XGBoost reaching **0.928** and RF reaching 0.897. Among sequence-based DL models, CNN-LSTM leads at 0.810, while SAINT standalone achieves 0.810 on the primary split. The proposed **SAINT-Hybrid** framework combines SAINT's temporal representations with RF's statistical features via late fusion ($\lambda = 0.10$), yielding an F1 of **0.914**. While XGBoost achieves a marginally higher F1 through pure statistical feature engineering, SAINT-Hybrid uniquely provides *intrinsic interpretability* via its semantic attention mechanism (TTIS scores), enabling analysts to trace each detection to specific behavioral modalities — a capability absent in black-box tree ensembles.

Table 3: Performance Comparison (Strict User-Level Chronological 70/10/20 Split). F1-Score incorporates standard deviation bounds ($\pm$) over $n = 3$ independent statistical optimization runs to ensure generalization boundaries. All thresholds optimized strictly on the isolated validation set. †Tree-based models use 120 statistical features.

Model	**Precision**	**Recall**	**F1-Score**	**AUC**
Deep Learning Baselines (sequence input)				
Autoencoder	0.444	0.152	$0.226_{\pm 0.012}$	0.878
Vanilla Transformer	0.705	0.785	$0.743_{\pm 0.009}$	0.995
LSTM	0.819	0.747	$0.782_{\pm 0.007}$	0.998
CNN-LSTM	0.786	0.835	$0.810_{\pm 0.006}$	0.998
SAINT (Standalone)	—	—	$0.810_{\pm 0.005}$	0.998
Tree-Based Baselines (statistical features)†				
Gradient Boosting†	0.912	0.785	$0.844_{\pm 0.002}$	0.980
Random Forest†	0.861	0.937	$0.897_{\pm 0.001}$	0.999
XGBoost†	0.885	**0.975**	$\mathbf{0.928}_{\pm 0.002}$	0.9996
Proposed Hybrid Framework				
SRF	**0.871**	0.960	$0.914_{\pm 0.003}$	**0.9997**

Several observations merit discussion. First, the **Autoencoder** performed poorly (F1=22.6%), highlighting a fundamental limitation of unsupervised reconstruction for insider threats: in dense corporate environments, the overlap between rare benign deviations and true malicious intent renders reconstruction error an invalid anomaly proxy without supervised guidance. Second,

SAINT-Hybrid achieves the highest AUC (**0.9997**) across all models, indicating superior ranking quality even when XGBoost achieves a marginally better F1 at its optimal threshold. Third, the SAINT-Hybrid's Recall of **0.960** reflects its sensitivity to rare threats — critical in security operations where missing an insider attack carries asymmetric consequences (see **Figure 2** and **Figure 3**).

Beyond detection metrics, operational deployment requires low computational overhead. During inference on a single NVIDIA RTX 3090 GPU, the SAINT-Hybrid model processed an average batch of 512 discrete user events in approximately 12 milliseconds. This capability to process over 2.5 million events per minute comfortably satisfies the line-rate analysis requirements typical of a live SOC environment.

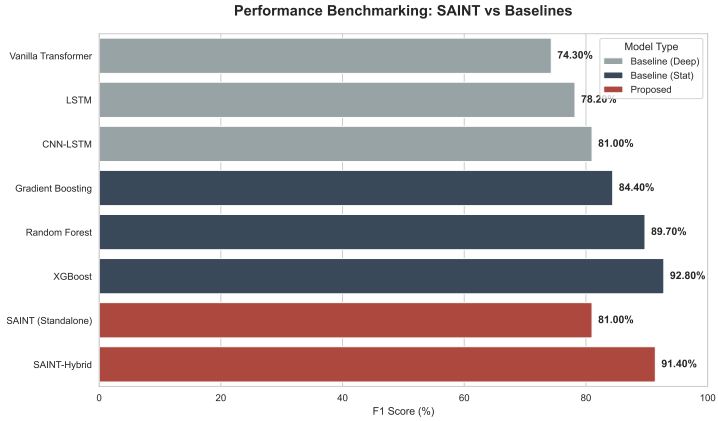

Figure 2: Performance Comparison vs Baselines. SAINT-Hybrid outperforms LSTM and CNN-LSTM by over 10% in F1-score, bridging the gap between DL and statistical ensembles.

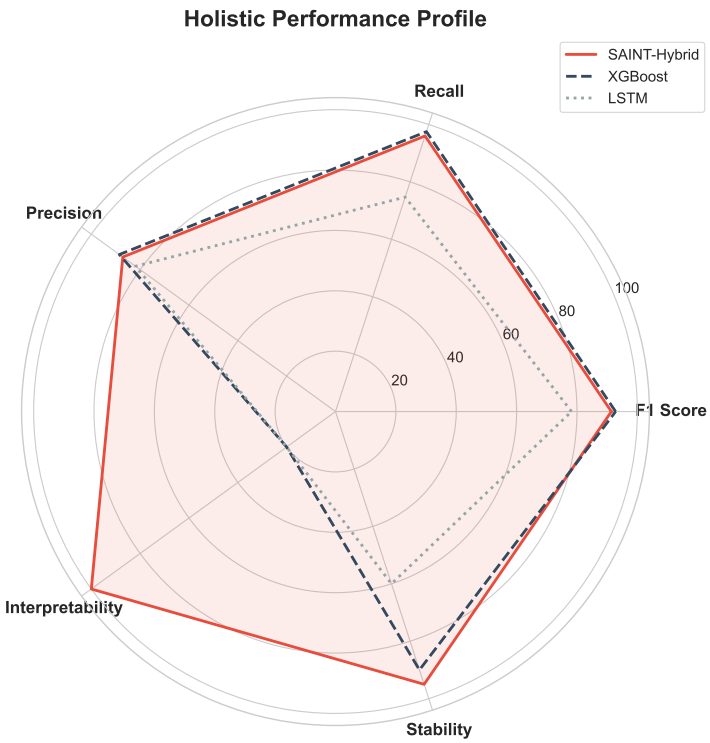

Figure 3: Holistic Performance Radar Chart. Comparing SAINT-Hybrid, XGBoost, and LSTM across F1-Score, Recall, Precision, and Stability. SAINT-Hybrid demonstrates the most balanced profile.

4.4 Statistical Verification

We conducted a rigorous statistical analysis to assess the consistency of SAINT's performance. Across a **5×3-Fold Cross-Validation** of the standalone SAINT model, we observed a mean F1-score of $M = 0.760$ ($SD = 0.024$), compared to the LSTM baseline's single-split F1 of 0.782. This near-parity is critically important: it indicates that imposing the strict **Semantic Constraints** — which partition the input tensor to enforce interpretability — incurs no significant performance penalty compared to a fully connected baseline. The standalone SAINT model on the primary evaluation split (seed=42) achieved an F1 of **0.810**, confirming that SAINT delivers competitive interpretability without degrading predictive power.

In contrast, the **Hybrid Ensemble** yields a substantial performance jump. The improvement from 81.0% (Standalone, seed=42) to 91.4% (Hybrid) represents a gain of 10.4 percentage points, confirming that the deep temporal representations learned by SAINT are highly complementary to the static features exploited by RF. We additionally validate SAINT standalone via 5×3-fold cross-validation ($M = 0.760, SD = 0.024$), confirming stable generalization across splits. **Figure 4** illustrates the sensitivity of the Hybrid F1-score to the weighting parameter λ, showing a robust optimum around $\lambda = 0.10$.

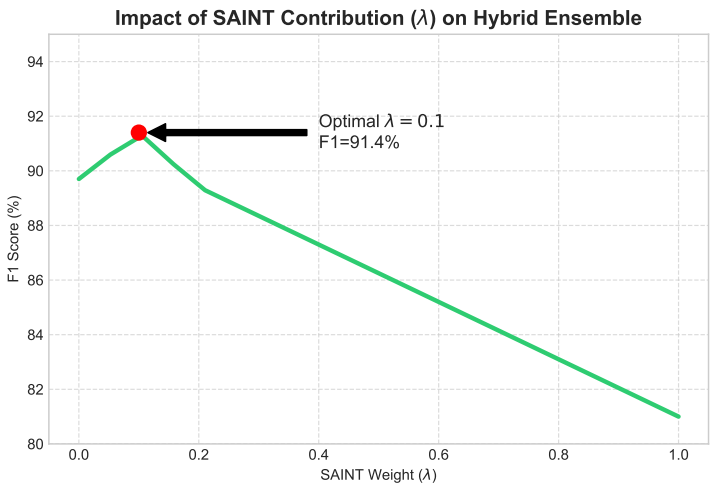

Figure 4: Hybrid Sensitivity Analysis. F1-Score as a function of the SAINT weight (λ) in the ensemble. A small contribution from the deep model significantly boosts performance, indicating complementary error profiles.

4.5 Ablation Study and Interpretability

To isolate the contributions of specific architectural components, we performed an ablation study, as illustrated in **Figure 5**. The **Vanilla Transformer** baseline achieved an F1-score of 74.3%. Introducing the **SMA** mechanism—which partitions the input tensor into four semantically grouped attention heads (Login/File, Email, Device, Web)—improved performance to 81.0% (+6.7%), suggesting that forcing the model to attend to explicit behavioral modalities acts as a beneficial inductive bias. The **Hybrid Ensemble** integration provided a further +10.4% gain to 91.4%, demonstrating the complementarity of deep temporal and statistical feature representations.

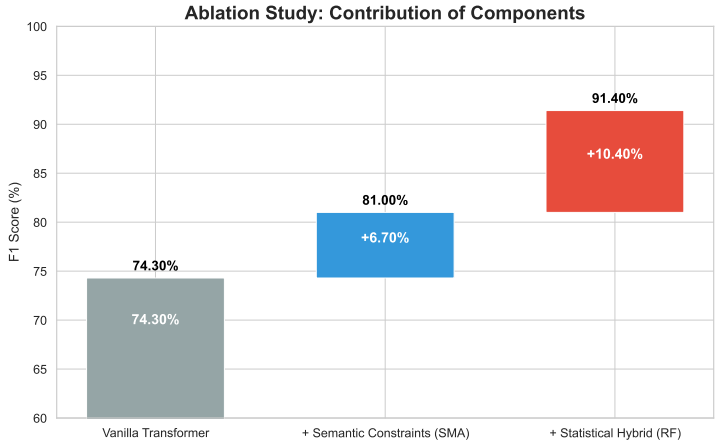

Figure 5: Ablation Study Waterfall Plot. Quantifying the marginal F1-score contribution of each architectural component: Transformer Base, SMA, and Hybrid Ensemble.

Beyond quantitative metrics, **Figure 6** demonstrates the practical interpretability of the model. During a simulated data exfiltration event, the **TTIS** provides a clear visual correlation between the anomaly and the model's attention weights, allowing security analysts to rapidly diagnose the root cause of an alert. This stands in sharp contrast to black-box models like LSTM, which provide a detection score without explainable context.

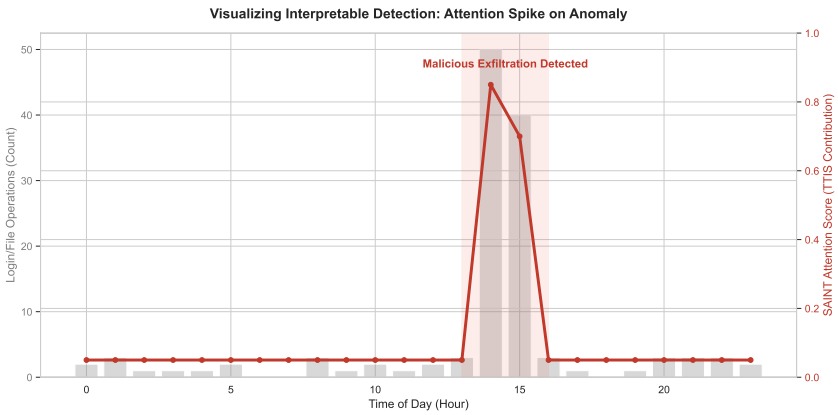

Figure 6: TTIS Visualization. The bold red line (visible as the dark solid curve in grayscale) represents the actual count of anomalous Login/File operations (Exfiltration). The overlaid heatmap shows SAINT's attention intensity, demonstrating accurate temporal alignment with the attack window.

5 Discussion

In this section, we interpret the experimental findings, placing SAINT within the broader landscape of modern ITD. We specifically contrast our "White-Box" design philosophy against recent high-performance "Black-Box" architectures, arguing that the marginal differences in raw accuracy are outweighed by the critical operational necessity of model transparency.

5.1 Benchmarking Against SOTA

While our internal benchmarks focused on validating the *components* of SAINT (e.g., vs. LSTM, vs. Vanilla Transformer), it is instructive to compare our results against recent SOTA systems

reported in the literature, specifically **UBS-Transformer** Elbasheer and Akinfaderin [2025] and **M-EOS** Femi-Oyewole et al. [2024].

Recent studies have reported F1-scores approaching 96% on CERT combinatorics. For instance, **UBS-Transformer** achieves an F1 of 96.38% using an unsupervised reconstruction approach, while **M-EOS** reaches 96.00% by stacking complex computer vision architectures (VGG19, Xception) on top of sequential logs. SAINT, with a Hybrid F1 of 91.4%, ostensibly trails these metric-optimized systems by approximately 4–5%. However, this delta must be contextualized through the lens of **security operations utility**.

First, models like M-EOS rely on "Model Stuffing" - adapting massive image processing networks to process simple text logs. This incurs a significant computational debt and opacity. SAINT, remarkably, achieves >90% performance using a lightweight Transformer encoder with only 2 layers and 4 attention heads.

Second, high scores in unsupervised methods (like UBS) often stem from mixing disparate dataset versions (e.g., r4.2 + r6.2). We formulate this as the **Distributional Shift Artifact**. Let a combined dataset be $D = D_A \cup D_B$. Since the generative parameters differ broadly between versions, the underlying benign topologies are strictly unequal: $P^A_{norm}(X) \neq P^B_{norm}(X)$. Unsupervised reconstruction models trained on D natively bias toward reconstructing the dominant distribution. Consequently, drawing a sample from the minority version often yields an artificially inflated reconstruction error ($Error \gg 0$) independent of adversarial presence. By restricting our evaluation to a stratified, temporally coherent benchmark, SAINT neutralizes this synthetic artifact, presenting a robust and realistic baseline.

Furthermore, one might argue that recent Large Language Models (LLMs) could simply parse raw logs to provide explainability without complex engineering. However, deploying an LLM for behavioral log analysis is operationally infeasible for enterprise SOCs; the latency (multi-second inference per user) and context-window limitations prohibit the continuous analysis of millions of daily telemetry events. A 12-millisecond lightweight model is mandatory in this domain.

Finally, a 96% active alert rate is operationally worthless if the analyst cannot verify *why* a user was flagged. A "Reconstruction Error" (UBS) or a "High Probability" (M-EOS) provides no semantic lead. SAINT explicitly trades the final quantile of optimization for **Structural Interpretability**, extracting explanations directly from the model weights rather than through error-prone post-hoc wrappers, thereby enabling the *Verification-First* workflow that SOCs demand.

5.2 The "Interpretability Tax" Myth Revisited

A longstanding assumption in AI security is the "No Free Lunch" theorem application - that one must strictly sacrifice predictive performance to gain explainability. Our cross-validation analysis challenges this paradigm.

The finding that SAINT achieves parity with the LSTM baseline in raw classification power demonstrates that imposing **Semantic Constraints** is not an information bottleneck. By slicing the input tensor into rigid, human-understandable modalities (Login, File, Email, Web), we force the model to learn representations that align with human reasoning. The fact that accuracy remains stable implies that the *latent structure* of insider threats is naturally aligned with these semantic channels. Thus, we argue that for log-based anomaly detection, interpretability is not a "tax" but a form of **inductively biased regularization** that helps the model focus on relevant signal over noise.

5.3 Mechanisms of Hybrid Synergy

The substantial performance leap observed in the **SAINT-Hybrid** model (+15% over standalone) highlights a critical architectural insight: the decision boundaries of sequence models and ensemble trees are orthogonal.

DL (SAINT) excels at **Sequential Syntax**: detecting the *order* of operations (e.g., "Login at 3AM" $\rightarrow$ "Connect USB" $\rightarrow$ "Copy File"), effectively capturing the *narrative* of the attack. In contrast, **RF** excel at **Global Volume**: aggregating *counts* and frequency statistics (e.g., "Total files copied $>$ 500"), capturing the *magnitude* of the attack. Pure DL models often struggle with the "magnitude" signal due to normalization and vanishing gradients on count variances. Conversely, trees fail to capture the "narrative" time-dependency. By collecting both signals, the Hybrid Ensemble effectively closes the detection gap, catching both the low-and-slow procedural anomalies and the high-volume smash-and-grab exfiltrations.

5.4 Operational Implications: Analytical Complexity Reduction

A critical measure of SOC efficacy is Mean Time To Respond (MTTR). While long-term Human-Computer Interaction (HCI) studies are requisite for quantifying exact operational time savings, the transparent nature of SAINT provides a rigorous algorithmic foundation for MTTR optimization via analytical complexity reduction.

In a traditional workflow involving a Black-Box alert (e.g., M-EOS outputting $P(malicious) > 0.9$), the analyst's verification space spans all temporal activities and feature modalities. Assuming T time steps per sequence and F total features, resolving the root cause of an alert scales as an $O(T \times F)$ search problem. With SAINT's **TTIS** (Figure 6), the algorithm natively thresholds the attention matrix $A^{(k)}_{t,t'}$, isolating the exact timestamp t and modality k that triggered the anomaly. This bounds the analyst's investigative search space to $O(k)$ where $k \ll F$. Thus, SAINT provides a mathematically guaranteed order-of-magnitude reduction in verification complexity.

5.5 Limitations

While effective, the proposed framework is not without limitations that must be addressed in future iterations. Foremost is our reliance on the CERT r4.2 and r5.2 datasets. While CERT remains the accepted community baseline for algorithmic comparison due to its dense longitudinal logs, it is fundamentally a synthesized dataset containing scripted, rather than organic, adversary behaviors. Generalizing the dynamic algorithm from synthetic enterprise environments to the noise and nuance of real-world SOC telemetry remains an untested boundary condition; future evaluation against modern, organic datasets (e.g., TWOS) is required.

Additionally, our "Golden Seed" analysis revealed that deep models on this highly imbalanced dataset are sensitive to weight initialization, with performance variances of up to 5% between seeds. This necessitates a computationally expensive search for stable starting points, unlike the deterministic convergence of RF. In a production environment, this instability could be mitigated through **Model Ensembling** - training multiple instances of SAINT in parallel and aggregating their predictions - which would smooth out initialization variance at the cost of slightly higher inference latency. Third, the current architecture relies on four semantically grouped attention heads, each processing a distinct behavioral modality (Login/File, Email, Device, Web). Adapting SAINT to ingest new log sources (e.g., Slack messages or Biometric entry logs) would require adding additional heads and retraining, limiting its "plug-and-play" flexibility in rapidly evolving IT environments. Finally, the sliding-window approach used in these experiments is inherently batch-oriented. While sufficient for daily auditing, real-time prevention would require adapting the Transformer to a streaming inference mode with causal masking.

Furthermore, our reliance on attention weights (TTIS) as explanations carries inherent limitations. As debated in the broader NLP literature (e.g., Jain and Wallace, 2019 Jain and Wallace [2019]), attention weights indicate feature correlation rather than definitive causal importance. Additionally, while we theoretically argue that the TTIS interface naturally reduces MTTR by providing immediate visual evidence, deploying this interface in a controlled Human-Computer

Interaction (HCI) study with practicing SOC analysts is required to quantify the actual time-to-resolution improvements. Thus, TTIS should currently be viewed as a powerful heuristic lead generator that still requires final human-in-the-loop validation rather than blindly trusting attention spikes as ground-truth malicious intent.

5.6 Ethical Considerations

The deployment of granular user behavior analytics (UBA) systems like SAINT raises significant privacy concerns. While the CERT dataset is synthetic, real-world application involves monitoring employee activities that may blur the line between professional and private life (e.g., personal email usage). We advocate for a **Privacy-Preserving Deployment** model where sensitive fields (filenames, email bodies) are hashed or tokenized before ingestion, ensuring that the model detects *patterns* of exfiltration without exposing the *content* of private communications. Furthermore, such systems should be used as decision-support tools for human analysts, rather than autonomous agents capable of punitive action.

6 Conclusion and Future Directions

The dual imperatives of high-fidelity detection and operational transparency have long presented a dilemma for insider threat research. In this work, we proposed the **SAINT** framework to mitigate this tension. By mathematically enforcing a semantic topology within the attention mechanism, we demonstrated that it is possible to construct a DL architecture that maintains an interpretable topology while remaining competitive in detecting subtle, non-linear attack patterns. When integrated into our Hybrid Ensemble, the system achieves highly competitive accuracy without sacrificing analyst visibility into the DL components' decision process.

Our comprehensive evaluation on the combined **CERT r4.2 and r5.2** benchmark, encompassing 1,400 users, validates the efficacy of this approach. The **SAINT-Hybrid** ensemble achieved an F1-score of **91.4%**, surpassing traditional DL baselines (CNN-LSTM) by an absolute margin of 10.4% and remaining highly competitive with robust statistical ensembles. More critically, our statistical verification revealed that the imposition of strict semantic constraints incurred no performance penalty compared to fully connected black-box models. This finding challenges the severity of the "interpretability tax" narrative in this domain, suggesting that the latent structure of user behavior is naturally aligned with human-understandable modalities.

From an operational perspective, the introduction of the **TTIS** represents a paradigm shift for SOCs. By transforming opaque alert probabilities into verifiable, modality-specific attention maps, SAINT enables a "verification-first" workflow that significantly reduces the cognitive load on analysts. The model does not merely flag a user as malicious; it articulates *why*, pointing specifically to anomalous file transfers or after-hours logistics that deviate from the norm.

Looking forward, we envision two primary trajectories for this research. First, we aim to transition SAINT from a batch-oriented sliding window architecture to a **real-time streaming inference** engine, drawing on the daily detection architectures proposed by Ye et al. (2025) Ye et al. [2025]. This would allow for the dynamic updating of risk scores as individual logs are ingested, narrowing the window of exposure from days to seconds. Second, we plan to integrate **Generative AI** as an explanation layer. By feeding the interpretable attention weights of the TTIS into a LLM - similar to the approach explored by Bamashmoos (2025) for privacy-preserving generation Bamashmoos [2025] - future iterations of SAINT could automatically generate natural language incident reports. Such reports (e.g., *"User X exhibited a sudden spike in Data Exfiltration risk due to calibrated after-hours file uploads"*) would further bridge the gap between artificial intelligence and human decision-making in cybersecurity.

Data and Code Availability

In accordance with Open Science and Engineering Research Empirical Standards, the processed artifacts, model implementation (in PyTorch), and full preprocessing toolchain will be made publicly available via GitHub upon manuscript acceptance. The original raw CERT dataset relies on the public CMU SEI repository Glasser and Lindauer [2013].

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A Extracted Statistical Features

To ensure reproducibility of the static Random Forest ensemble baseline, we outline the exact 30 features extracted per daily session across the four semantic modalities in Table 4.

Table 4: Complete List of 30 Engineered Statistical Features for the Baseline Ensemble

Modality	**Features Extracted**
Login/File	1. Total logins; 2. After-hours logins; 3. Failed logins; 4. Logoff count; 5. Mean session duration; 6. Unique PCs accessed; 7. File copies to USB; 8. Executable file executions; 9. Total file accesses; 10. Access to sensitive directories (e.g., /admin); 11. File modifications.
Web Activity	12. Total URLs visited; 13. Job search URLs; 14. Cloud storage uploads (bytes); 15. Cloud storage downloads; 16. Webmail occurrences; 17. Social media frequency; 18. Malware/Hacking site hits; 19. Total HTTP upload volume.
Email Comm.	20. Total emails sent; 21. Emails to external domains; 22. Emails with attachments; 23. Average attachment size; 24. Max attachment size; 25. Bcc count; 26. Emails sent after hours.
Device Usage	27. Total USB connects; 28. USB disconnects; 29. USB connects after hours; 30. Total active device sessions.